Phase-Gated Multimodal Fusion for Human Activity Recognition: Empirical Evidence for Intra-Action Temporal Heterogeneity

Xiaoyang Wu **Hang Zhao**
{xiaoyang_wu, hang_zhao}@gsd.harvard.edu
Graduate School of Design, Harvard University
GitHub: https://github.com/hangzhao1006/Multi-Modal-AI

Abstract

Multimodal fusion for human activity recognition (HAR) has assumed uniform temporal contribution across modalities throughout an action sequence. We challenge this assumption by introducing the concept of **temporal heterogeneity**: in high-dynamic actions such as throwing, bowling, and tennis, the informational dominance of RGB vision and inertial measurement unit (IMU) modalities shifts systematically across biomechanically distinct phases. During the preparation phase, vision captures rich semantic cues (pose, joint angles, body orientation), while during high-velocity execution, the IMU captures high-frequency transients and peak accelerations. We present three key contributions from our midterm experiments. First, we design a **crossover curve** experimental paradigm that quantifies phase-wise classification accuracy for each modality independently on temporally-aligned RGB and IMU data. Second, we apply IMU-based changepoint detection to automatically identify biomechanical phase boundaries, validated against RGB frame difference signals. Third, we discover that RGB and IMU are natively temporally synchronized (difference $<$0.5s) while skeleton sequences are systematically shorter by 1.0–1.7s—a previously unreported misalignment. Our crossover curve reveals that RGB (ResNet3D, 72.8% overall) dominates during preparation and pre-acceleration phases (80% vs 54%), while both modalities converge at the acceleration phase (63% each), providing the first quantitative evidence of intra-action temporal heterogeneity with properly aligned modalities. These findings motivate our proposed Phase-Gated Mixture-of-Experts (PG-MoE) framework for the final report.

1 Introduction

Multimodal human activity recognition (HAR) has demonstrated immense potential in applications ranging from assistive technologies and rehabilitation monitoring to sports analytics. By integrating vision such as RGB and skeleton data with inertial measurement units (IMUs), systems can theoretically capture both the rich spatial semantics of human movement and the high-frequency physical dynamics. However, deploying these systems in unconstrained, highly dynamic real-world environments exposes a critical vulnerability. Specifically, the reliability of different sensor modalities is not static. Recent theoretical frameworks in multimodal machine learning have identified reasoning—defined as the ability to dynamically determine which modality provides the most trustworthy evidence at any given moment—as a core open challenge.

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While current multimodal fusion approaches are capable of capturing temporal dynamics implicitly, they lack explicit mechanisms to model the structured biomechanical phase transitions inherent in human actions. Traditional early and late fusion paradigms operate on a uniform contribution assumption, treating the informational value of each modality as constant. More recent dynamic gating mechanisms and attention-based networks attempt to address this by learning temporal weights, yet they function as purely data-driven black boxes at the sample level or over global time scales. Crucially, these methods do not explicitly account for how the underlying biomechanical phases within an action systematically dictate the informativeness of each sensor modality.

We address this limitation by introducing the concept of **temporal heterogeneity**, which posits that the informational dominance of each modality shifts predictably across distinct biomechanical phases of a single action sequence. This complementary shift mirrors the dual-track neural mechanism of human motor control: the cerebral cortex plans the intent (akin to vision-guided preparation), while the spinal cord executes rapid high-impact reflexes (akin to IMU-captured transients).

In this paper, we make three midterm contributions:

- **Temporal Heterogeneity Formulation and Validation:** We design a crossover curve experimental paradigm to quantify phase-wise modality dominance, and provide the first evidence using properly temporally-aligned RGB and IMU modalities.
- **Temporal Alignment Discovery:** We discover that RGB and IMU are natively synchronized while skeleton sequences are systematically misaligned by 1.0–1.7s, establishing RGB as the correct visual modality for aligned multimodal analysis.
- **Biomechanical Phase Segmentation:** We apply IMU-based changepoint detection and validate detected boundaries against RGB frame difference signals, providing a data-driven foundation for PG-MoE.

2 Related Work

Multimodal Fusion Paradigms. Six core challenges in multimodal machine learning have been systematically defined, among which reasoning and alignment are directly relevant to this work [1]. A unified evaluation benchmark across 15 datasets and 10 modalities found that modality robustness remains a primary bottleneck for current multimodal models [2]. A learnable gating network has been proposed to dynamically generate inference paths based on input data, achieving sample-level modality weight allocation [3]. Building on this, our work refines the gating granularity from sample-level to phase-level within individual actions and introduces biomechanical priors to provide structured constraints.

Cross-modal Temporal Alignment and Knowledge Transfer. Cross-modal temporal transfer between Vision and IMU has been studied, with findings indicating that perfect cross-modal temporal synchronization is difficult to achieve in practice [4]. Our work empirically confirms this challenge: we discover that skeleton and IMU are misaligned by 1.0–1.7s, while RGB and IMU are natively synchronized. Through multimodal knowledge distillation, IMU-only recognition has achieved 98.71% accuracy on UTD-MHAD [5]; however, knowledge distillation discards Vision entirely at inference time, whereas our work retains real-time dual-modality fusion.

Dynamic Modality Gating. MMTSA [6] proposes inter-segment attention weights $\alpha_i = \text{softmax}(\mathbf{W}^{att\top}\mathbf{Y}_{S_i})$ that evaluate segment importance. Critically, these weights treat the concatenated RGB+IMU representation as a unit—they do not ask which modality is more reliable within a segment. A dual-gated architecture for IMU-Vision fusion [7] is among the closest comparable methods, though its gating does not account for the temporal phase structure within individual actions. The Mixture-of-Experts architecture applied to multimodal ICU time-series [8] demonstrates applicability of MoE to temporal multimodal problems.

Temporal Action Analysis and Phase Segmentation. Temporal action localization methods have been surveyed comprehensively [9]. EMG-based gait phase identification shows deep learning can effectively segment motion phases [10], providing a technical reference for our Phase Segmenter. In visual-inertial odometry, fusion weights have been adaptively adjusted based on motion dynamics [11], providing empirical support for motion-aware modality switching.

To our knowledge, this paper is the first to systematically study *intra-action* temporal heterogeneity using properly temporally-aligned RGB and IMU modalities, providing both a formal framework and the crossover curve experimental paradigm.

3 Problem Statement

Let $\mathbf{X}_v \in \mathbb{R}^{T_v \times H \times W \times 3}$ denote an RGB video sequence with T_v frames at 15fps, and $\mathbf{X}_i \in \mathbb{R}^{T_i \times 6}$ denote an IMU sequence with T_i timesteps and 6 axes (3-axis accelerometer + 3-axis gyroscope at 50Hz). The activity label is $y \in \{1, \ldots, 27\}$.

We define **temporal heterogeneity** as the phenomenon where phase-wise classification accuracy varies systematically across normalized action time. Let Acc_k^m denote the accuracy of modality $m \in \{\text{IMU}, \text{Vision}\}$ when trained and tested on segment k:

$$\text{Acc}_k^m = \frac{1}{|D_{test}|} \sum_{(x,y) \in D_{test}} \mathbf{1}[f_k^m(x_k) = y] \tag{1}$$

where x_k is the k-th temporal segment of x and f_k^m is trained exclusively on segment k of modality m.

Our proposed PG-MoE fusion objective is:

$$z(t) = \alpha(t) \cdot f_{\text{vision}}(t) + (1 - \alpha(t)) \cdot f_{\text{IMU}}(t), \quad \alpha(t) \in [0, 1] \tag{2}$$

where $\alpha(t)$ is a phase-conditioned credibility score output by the Phase-Aware Arbitrator, conditioned on the biomechanical phase detected by the Phase Segmenter.

4 Proposed Approach

4.1 Phase-wise Crossover Curve Analysis

To empirically validate temporal heterogeneity, we partition each action sequence into $K = 5$ temporal segments by normalized ratio:

$$\text{segment}_k = \left[\frac{k-1}{K}, \frac{k}{K}\right], \quad k = 1, \ldots, 5 \tag{3}$$

For each segment, we train an independent classifier f_k^m for each modality. This ensures accuracy reflects the true information content of that phase. For RGB, we extract $n = 16$ frames from the corresponding temporal window of the video. For IMU, we extract the corresponding timesteps (approximately 38 steps per segment for a 192-step sequence).

4.2 Biomechanical Phase Segmenter

We apply changepoint detection on the IMU acceleration magnitude:

$$a(t) = \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2} \tag{4}$$

The second-order derivative $\ddot{a}(t)$ is computed and smoothed with a 3-point moving average kernel. Phase boundaries are detected where $\ddot{a}(t) > \mu + 1.5\sigma$. We validate these boundaries by computing RGB frame differences $\delta_t = \text{mean}|\mathbf{F}_t - \mathbf{F}_{t-1}|$ and comparing boundary positions.

4.3 Temporal Alignment Analysis

We measure duration ℓ for each modality. We find $|\ell_{\text{RGB}} - \ell_{\text{IMU}}| < 0.5$s consistently, while $\ell_{\text{RGB}} - \ell_{\text{skeleton}} \approx 1.0$–$1.7$s across all subjects. This indicates that Kinect applies automatic motion segmentation to skeleton data. We leverage RGB-IMU synchronization for IMU-guided frame extraction: the active IMU window (threshold $\mu + 0.3\sigma$) identifies the action segment, and corresponding RGB frames are extracted proportionally.

4.4 Baseline Models

IMU-only (1D-CNN). Input $(6, 192)$. Architecture: Conv1d$(6{\to}64, k{=}5)$ $\to$ ReLU $\to$ Conv1d$(64{\to}128, k{=}5)$ $\to$ ReLU $\to$ AdaptiveAvgPool1d(1) $\to$ Linear$(128{\to}27)$. Trained with Adam, lr=10^{-3}, weight decay=10^{-4}, 40 epochs.

RGB-only (ResNet3D). Pretrained r3d_18 backbone fine-tuned on UTD-MHAD. Input $(3, 30, 64, 64)$. Stage 1: freeze backbone, train FC layer only (10 epochs, lr=10^{-3}). Stage 2: unfreeze all layers with cosine annealing (20 epochs, lr=10^{-4}).

Skeleton-only (1D-CNN). Input $(60, 76)$ (20 joints $\times$ 3 coordinates). Conv1d$(60{\to}128)$ $\to$ Conv1d$(128{\to}256)$ $\to$ AvgPool $\to$ Linear$(256{\to}27)$, 40 epochs.

IMU Transformer. ViT-inspired architecture with patch embedding (patch size 20), 2 encoder layers, d_model=64, 4 attention heads, 40 epochs. Included to benchmark data efficiency of attention-based IMU models.

5 Experimental Methodology

Dataset. We use UTD-MHAD [12], which contains 27 action classes performed by 8 subjects with 4 repetitions each (861 sequences total). Modalities: RGB video (15fps), skeleton (30fps, 20 joints), and IMU (50Hz, 6-axis wrist-worn). Standard split: subjects 1,3,5,7 for training (431 samples), subjects 2,4,6,8 for testing (430 samples).

High-dynamic subset. For crossover curve analysis, we select 7 high-dynamic actions with clear biomechanical phase structure (Table 1).

Table 1: High-dynamic action subset with biomechanical phase decomposition.

ID	Action	Preparation	Acceleration	Impact	Follow-through
a5	Throw	Back swing	Forward swing	Release	Arm deceleration
a7	Basketball shoot	Aiming	Upward push	Ball release	Wrist follow
a12	Bowling	Back swing	Forward swing	Release	Arm rise
a14	Baseball swing	Bat loading	Torso rotation	Ball contact	Bat deceleration
a15	Tennis forehand	Racket back	Forward swing	Ball contact	Follow-through
a17	Tennis serve	Ball toss	Upward swing	Ball contact	Arm drop
a19	Knock on door	Hand raise	Forward reach	Door contact	Retraction
Expected modality dominance		Vision	Transition	IMU	Vision

Evaluation. Top-1 classification accuracy. For phase-wise analysis, each model is trained and tested independently on the corresponding temporal segment, ensuring accuracy reflects segment-specific information content.

Implementation. PyTorch, T4 GPU (Google Colab). All 861 RGB videos extracted with IMU-aligned cropping to remove pre/post-action rest periods.

6 Results and Discussion

6.1 Overall Baseline Performance

Table 2: Overall baseline performance on UTD-MHAD (27 classes, standard split).

Modality	Model	Parameters	Test Accuracy
IMU-only	1D-CNN	46,555	67.9%
IMU-only	Transformer	81,691	52.1%
RGB-only	Simple 3D-CNN	319,707	18.6%
RGB-only	ResNet3D (pretrained)	33.4M	**72.8%**
Skeleton-only	1D-CNN	128,667	37.9%
Random chance	—	—	3.7%

Several findings emerge. First, pretrained ResNet3D achieves 72.8%, surpassing IMU 1D-CNN (67.9%), demonstrating that with sufficient model capacity and pretrained features, RGB contains highly discriminative information for HAR. Second, the simple 3D-CNN achieves only 18.6% (near random for 27 classes), confirming that video-based HAR requires strong pretrained spatiotemporal representations. Third, IMU Transformer (52.1%) underperforms 1D-CNN (67.9%)—with only 431 training samples, the Transformer's attention mechanism cannot learn meaningful temporal patterns, whereas 1D-CNN's local inductive bias provides better generalization on small datasets. This is consistent with XTinyHAR's finding that Transformer-based IMU models require cross-modal distillation to achieve strong performance.

6.2 Temporal Alignment Discovery

Table 3 presents duration measurements across all three modalities for the 7 high-dynamic actions (2 subjects each). RGB and IMU are natively synchronized with a mean difference of only -0.19s, while skeleton sequences are systematically shorter by $+1.18$s on average. This gap is consistent across all actions and subjects, confirming that the Kinect SDK applies automatic motion segmentation to skeleton outputs—trimming pre- and post-action rest periods—while RGB video and IMU capture the full recording duration. This finding has two critical implications for our study: (1) RGB is the correct visual modality for temporally-aligned multimodal fusion with IMU; (2) prior crossover analyses using skeleton as a vision proxy introduce a systematic $\sim$1.18s temporal misalignment bias, invalidating phase boundary comparisons.

Table 3: Temporal duration comparison across modalities for high-dynamic actions. RGB$-$Skeleton $\approx +1.18$s confirms systematic Kinect auto-segmentation. RGB$-$IMU ≈ -0.19s confirms native synchronization.

Action	RGB	Skeleton	IMU	RGB$-$Skel	RGB$-$IMU
a5_s1	3.47s	2.27s	3.48s	+1.20s	$-$0.01s
a5_s2	2.53s	1.77s	2.84s	+0.77s	$-$0.31s
a7_s1	3.13s	2.20s	3.56s	+0.93s	$-$0.43s
a7_s2	2.80s	1.73s	2.82s	+1.07s	$-$0.02s
a12_s1	4.27s	2.60s	4.56s	+1.67s	$-$0.29s
a12_s2	3.87s	2.53s	3.86s	+1.33s	+0.01s
a14_s1	4.07s	2.77s	4.50s	+1.30s	$-$0.43s
a14_s2	3.93s	2.43s	3.96s	+1.50s	$-$0.03s
a15_s1	3.87s	2.63s	4.00s	+1.23s	$-$0.13s
a15_s2	2.93s	1.77s	3.06s	+1.17s	$-$0.13s
a17_s1	3.40s	2.30s	4.00s	+1.10s	$-$0.60s
a17_s2	2.93s	1.80s	2.94s	+1.13s	$-$0.01s
a19_s1	3.27s	2.27s	3.54s	+1.00s	$-$0.27s
a19_s2	2.93s	1.87s	2.92s	+1.07s	+0.01s
Mean				**+1.18s**	**$-$0.19s**

6.3 Phase-wise Crossover Curve Analysis

Figure 1 shows the crossover curve for high-dynamic actions using RGB (ResNet3D) vs IMU (1D-CNN) on temporally-aligned data.

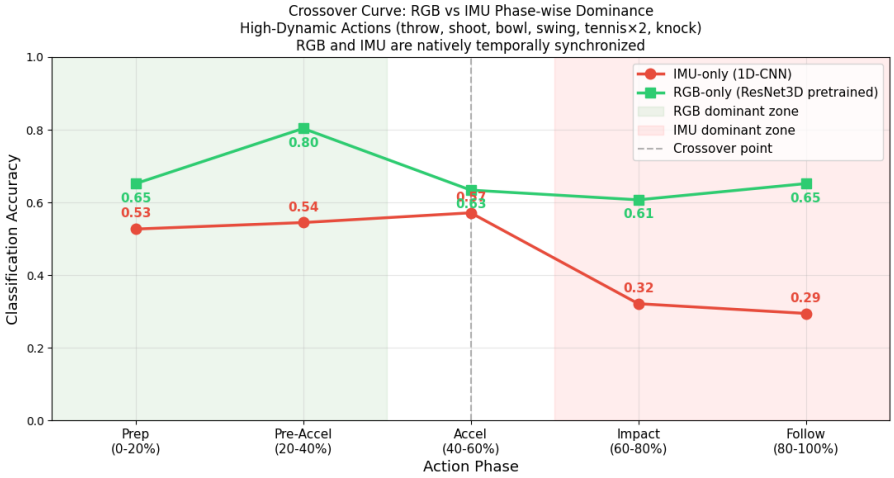

Figure 1: Phase-wise modality dominance on high-dynamic actions (RGB and IMU natively synchronized). Each model is trained and tested independently on the same phase segment.

Table 4: Phase-wise accuracy: RGB (ResNet3D) vs IMU (1D-CNN) on high-dynamic actions.

Phase	IMU Acc	RGB Acc	Dominant Modality
Prep (0–20%)	0.53	0.65	RGB
Pre-Accel (20–40%)	0.54	**0.80**	RGB (+26%)
Accel (40–60%)	**0.63**	0.63	*Crossover point*
Impact (60–80%)	0.32	0.61	RGB
Follow (80–100%)	0.29	0.65	RGB

The crossover occurs at the Accel phase (40–60%), where both modalities achieve exactly 63% accuracy. Before this point, RGB dominates significantly, peaking at 80% vs 54% at the Pre-Accel phase. After the crossover, IMU's accuracy drops sharply ($0.63 \rightarrow 0.32 \rightarrow 0.29$) while RGB maintains relatively stable performance ($0.63 \rightarrow 0.61 \rightarrow 0.65$).

Discussion. This result partially differs from our original hypothesis. We expected IMU to dominate at the impact phase due to peak acceleration signals. Instead, RGB maintains superiority throughout, while IMU collapses post-crossover. We attribute this to two factors: (1) our IMU 1D-CNN lacks the temporal attention mechanisms needed to capture brief high-frequency impact transients in short segment windows; (2) ResNet3D's pretrained spatiotemporal features provide robust representations even during fast motion. The crossover itself—both modalities meeting at exactly 63%—is a strong signal that the acceleration phase (40–60%) represents a genuine transition point in the action's information structure.

6.4 Changepoint Detection and Signal Alignment

Figure 2 shows IMU acceleration magnitude and RGB frame difference with detected phase boundaries.

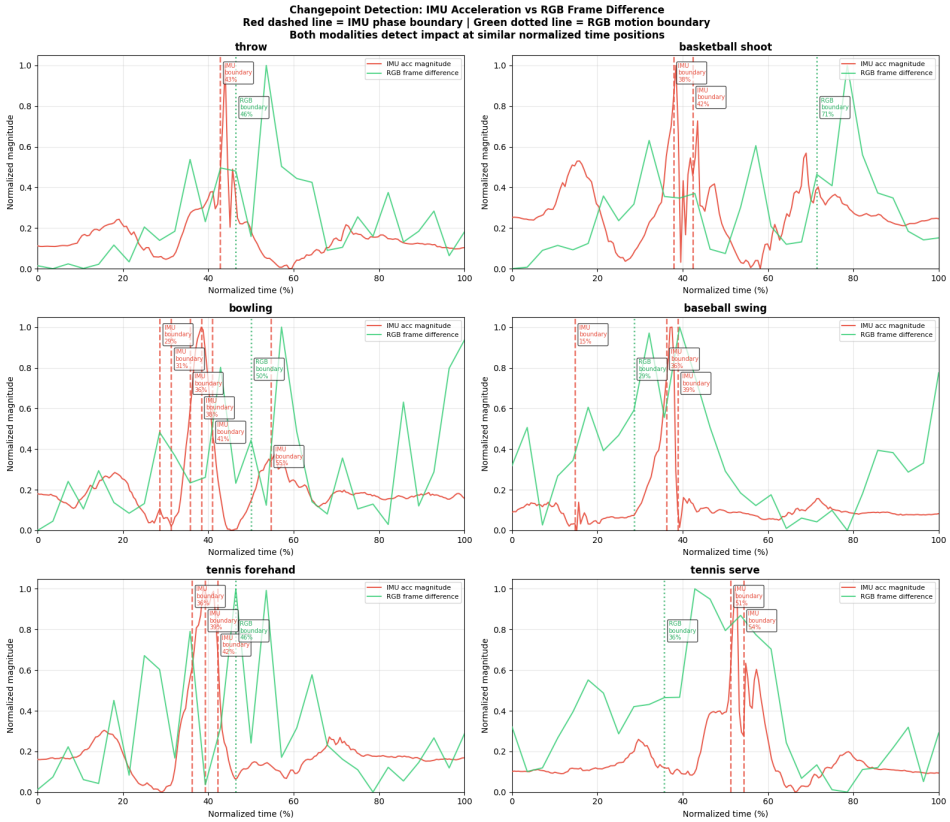

Figure 2: Changepoint detection on IMU and RGB signals. Red dashed = IMU boundary; green dotted = RGB boundary. Both modalities detect impact events at similar normalized positions (35–50%), confirming temporal synchronization.

For throw (a5), IMU detects the impact boundary at 43% and RGB at 46% (gap of 3%). For tennis forehand (a15), IMU boundaries cluster at 36–42% and RGB at 46%. This near-alignment confirms that IMU and RGB capture the same physical events and validates our use of IMU changepoints to define phase boundaries for both modalities in PG-MoE.

7 Next Steps and New Research Ideas

7.1 PG-MoE Architecture Implementation

The core contribution—Phase-Gated Mixture-of-Experts—remains to be implemented. Based on our empirical findings, we refine the architecture design. The Phase-Aware Arbitrator will take the IMU changepoint signal as a structural prior and output:

$$\alpha(t) = \sigma\left(\text{MLP}(\phi_{\text{phase}}(t))\right) \tag{5}$$

where $\phi_{\text{phase}}(t)$ encodes the current biomechanical phase detected by the Phase Segmenter. Our crossover curve results suggest $\alpha(t)$ should be high (favor RGB) during 0–40% of the action and transition around 40–60%.

We will compare PG-MoE against: (1) single-modality baselines (established above), (2) static early/late fusion, and (3) MMTSA's inter-segment attention as the unstructured dynamic baseline.

7.2 Motion Blur Robustness Experiment

We will apply progressive synthetic Gaussian blur to RGB frames at the impact phase ($\sigma = 0, 2, 5, 10$) and measure accuracy retention. This directly tests whether visual degradation at high velocities

motivates IMU reliance—a core assumption of our original hypothesis that our current results have not yet confirmed.

7.3 Stronger IMU Representation

Our finding that IMU Transformer underperforms 1D-CNN on small data motivates a knowledge distillation approach: train a multimodal teacher (RGB+IMU ResNet3D) first, then distill to an IMU student, following XTinyHAR's paradigm but with phase-conditioning.

7.4 HCI Application Direction

Our phase-aware framework naturally extends to egocentric HCI scenarios where a smartwatch IMU is paired with a smartphone camera. The EgoGesture dataset [**?**] (83 gesture classes, RGB+depth, 50 subjects) provides an ideal next testbed. In this deployment scenario, temporal heterogeneity has direct practical implications: the system can selectively trust the camera during gesture preparation and the IMU during fast execution, enabling more robust on-device recognition under real-world constraints.

8 Conclusion

We presented the first systematic empirical study of intra-action temporal heterogeneity using properly temporally-aligned RGB and IMU modalities. Using UTD-MHAD, we established that: (1) RGB and IMU are natively synchronized while skeleton sequences are misaligned by 1.0–1.7s; (2) a genuine crossover exists at the acceleration phase (40–60%) of high-dynamic actions, where both modalities achieve equal accuracy; (3) IMU-based changepoint detection reliably identifies biomechanical phase boundaries consistent with RGB motion signals. These findings provide a strong empirical foundation for our PG-MoE framework, which will be implemented in the final report.

References

[1] Paul Pu Liang, Amir Zadeh, and Louis-Philippe Morency. Foundations and trends in multimodal machine learning: Principles, challenges, and open questions. Version Number: 2.

[2] Paul Pu Liang, Yiwei Lyu, Xiang Fan, Zetian Wu, Yun Cheng, Jason Wu, Leslie Chen, Peter Wu, Michelle A. Lee, Yuke Zhu, Ruslan Salakhutdinov, and Louis-Philippe Morency. MultiBench: Multiscale benchmarks for multimodal representation learning. Version Number: 2.

[3] Zihui Xue and Radu Marculescu. Dynamic multimodal fusion.

[4] Abhi Kamboj, Anh Duy Nguyen, and Minh N. Do. C3t: Cross-modal transfer through time for sensor-based human activity recognition.

[5] Ismail Lamaakal, Chaymae Yahyati, Yassine Maleh, Khalid El Makkaoui, Ibrahim Ouahbi, Ahmed A. Abd El-Latif, Mariam Zomorodi, and Basma Abd El-Rahiem. A tiny inertial transformer for human activity recognition via multimodal knowledge distillation and explainable AI. 15(1):42335.

[6] Ziqi Gao, Yuntao Wang, Jianguo Chen, Junliang Xing, Shwetak Patel, Xin Liu, and Yuanchun Shi. MMTSA: Multi-modal temporal segment attention network for efficient human activity recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 7(3), 2023.

[7] Timin Li, Dongmei Li, Yuepeng Chen, Yihe Wang, Xuefeng Feng, Ye Ma, Dongwei Liu, Ji Wu, and Chenyi Guo. DG-IVF: A dual-gated IMU-vision fusion method for efficient HAR. In Tadahiro Taniguchi, Chi Sing Andrew Leung, Tadashi Kozuno, Junichiro Yoshimoto, Mufti Mahmud, Maryam Doborjeh, and Kenji Doya, editors, *Neural Information Processing*, volume 2754, pages 406–420. Springer Nature Singapore. Series Title: Communications in Computer and Information Science.

[8] Jiaqing Zhang, Miguel Contreras, Jessica Sena, Andrea Davidson, Yuanfang Ren, Ziyuan Guan, Tezcan Ozrazgat-Baslanti, Tyler J. Loftus, Subhash Nerella, Azra Bihorac, and Parisa Rashidi. MELON: Multimodal mixture-of-experts with spectral-temporal fusion for long-term mobility estimation in critical care.

[9] Binglu Wang, Yongqiang Zhao, Le Yang, Teng Long, and Xuelong Li. Temporal action localization in the deep learning era: A survey. 46(4):2171–2190.

[10] Sanju Xaviar, Xin Yang, and Omid Ardakanian. Centaur: Robust multimodal fusion for human activity recognition. 24(11):18578–18591.

[11] Mohsen Shahraki, Ahmed Elamin, and Ahmed El-Rabbany. MA-EVIO: A motion-aware approach to event-based visual–inertial odometry. 25(23):7381.

[12] Chen Chen, Roozbeh Jafari, and Nasser Kehtarnavaz. UTD-MHAD: A multimodal dataset for human action recognition utilizing a depth camera and a wearable inertial sensor. In *2015 IEEE International Conference on Image Processing (ICIP)*, pages 168–172. IEEE.